$$h_t = f_\phi(h_{t-1}, z_{t-1}, a_{t-1})$$

$$z_t \sim q_\phi(z_t \mid h_t, x_t)$$

$$\hat{z}_t \sim p_\phi(\hat{z}_t \mid h_t), \qquad s_t = [h_t, z_t]$$

$$\hat{x}_t, \hat{r}_t, \hat{c}_t \leftarrow p_\phi(\cdot \mid s_t)$$